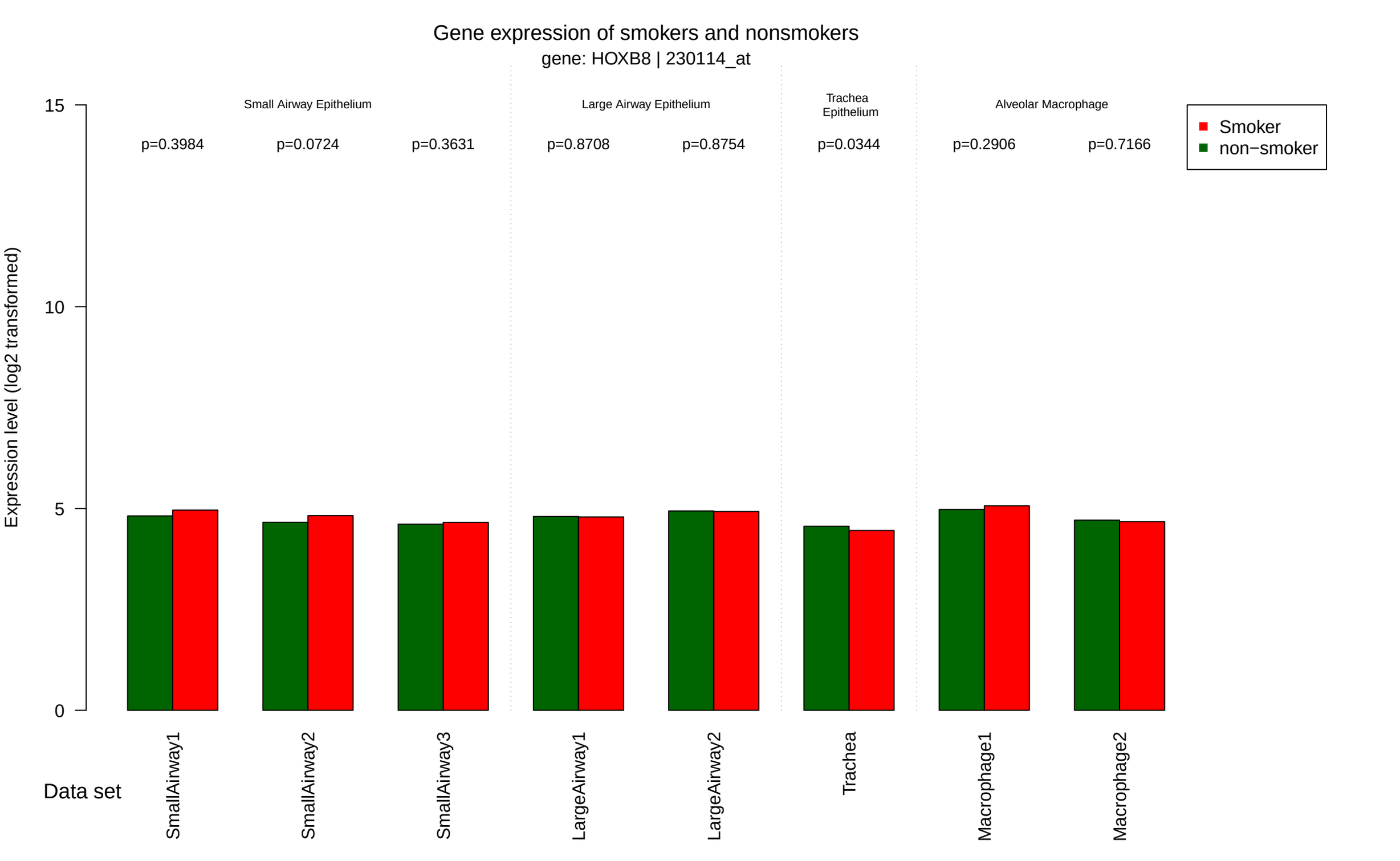
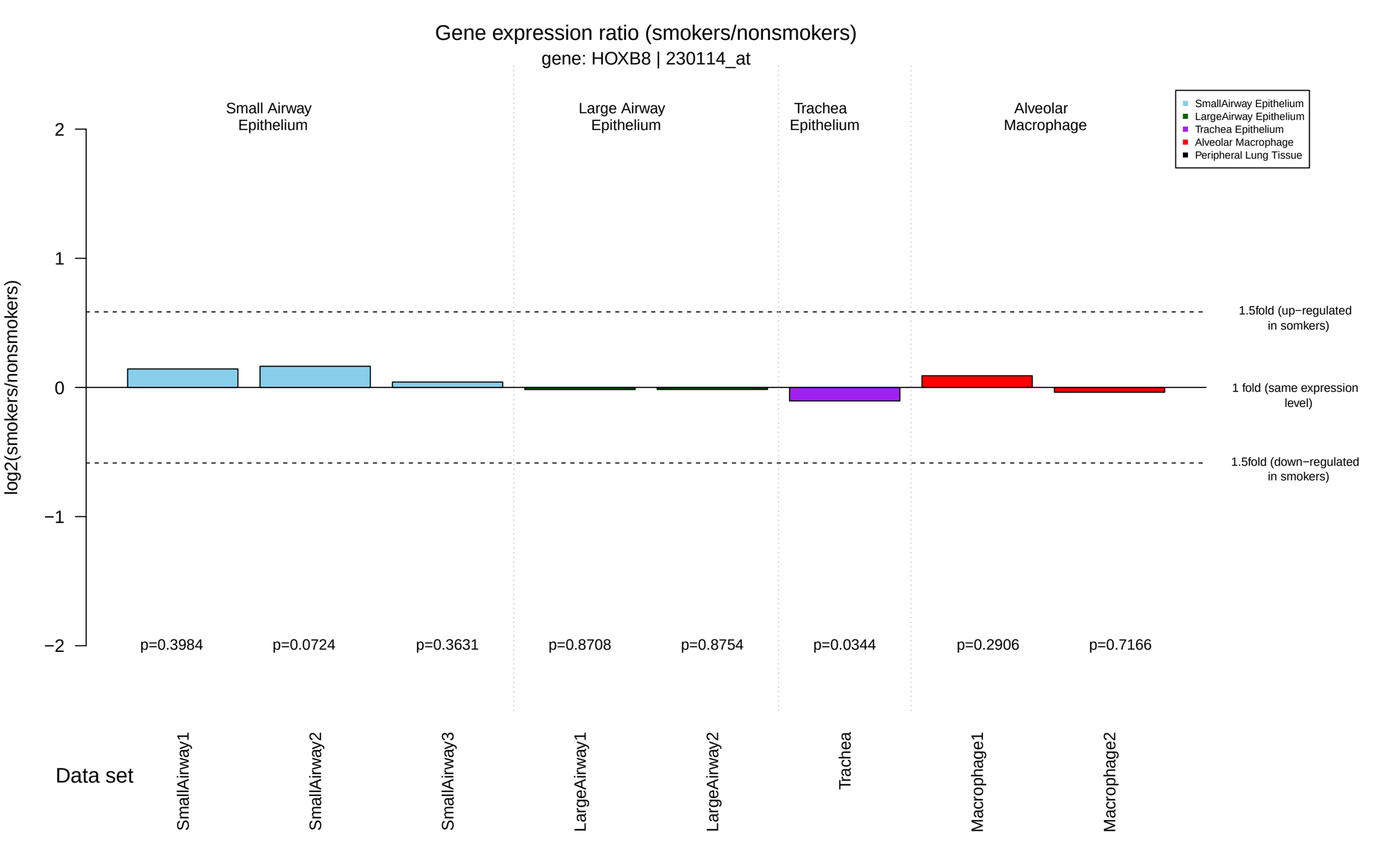
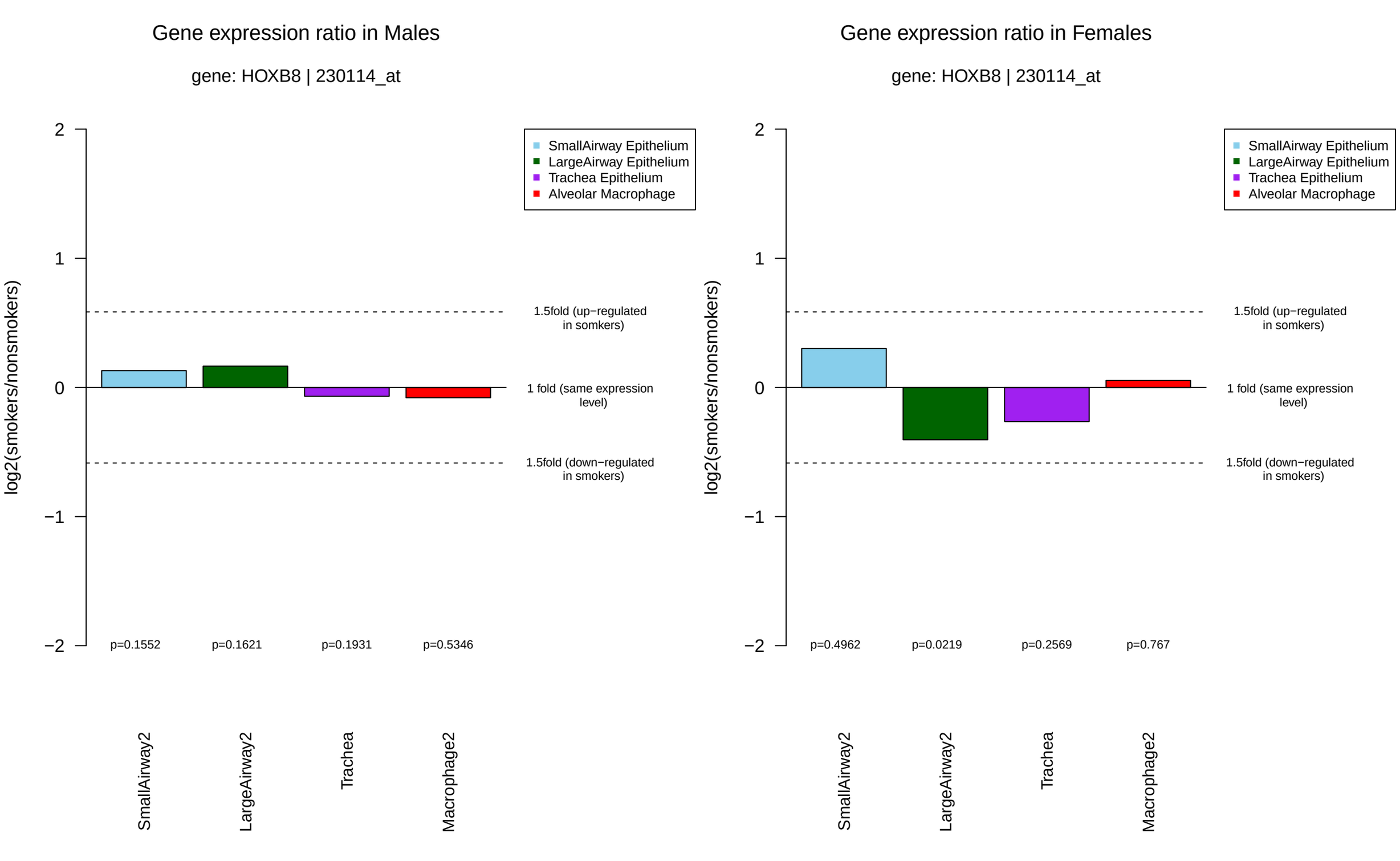
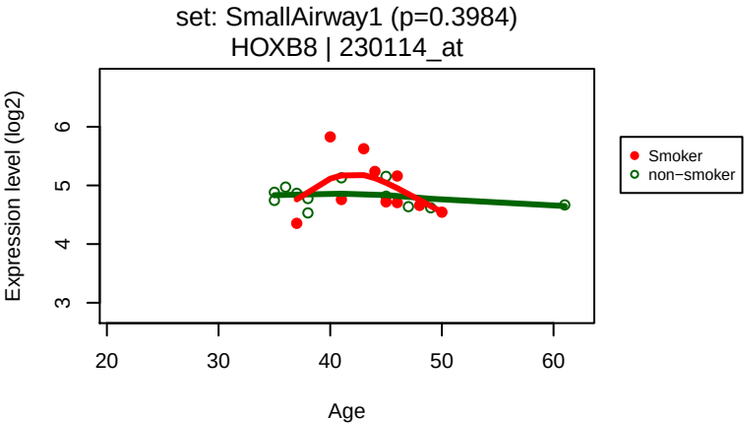
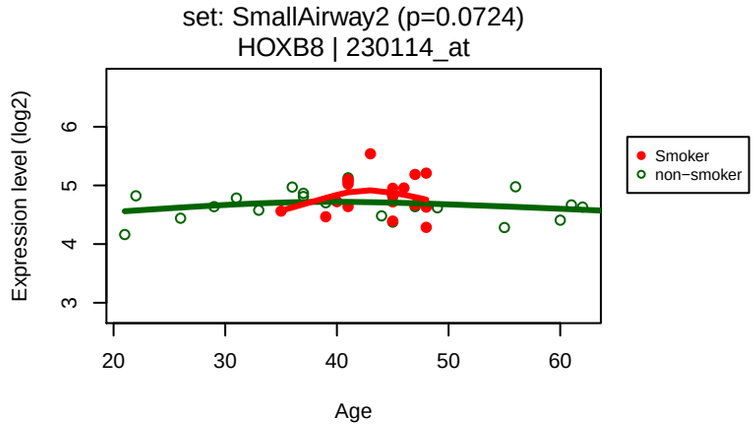
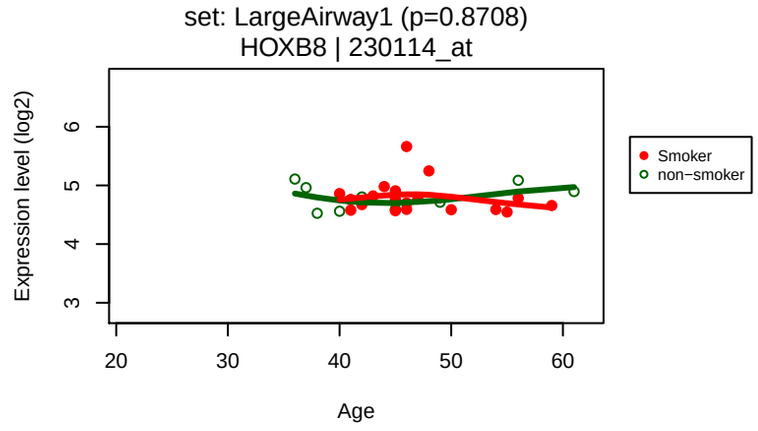

Gene expression of smokers and nonsmokers
gene: HOXB8 | 230114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3984
p=0.0724
p=0.3631
p=0.8708
p=0.8754
p=0.0344
p=0.2906
p=0.7166
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

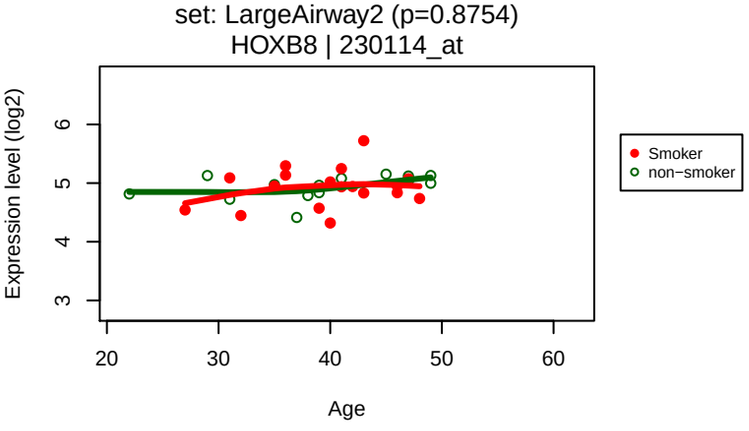

Gene expression ratio (smokers/nonsmokers)
gene: HOXB8 | 230114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3984
p=0.0724
p=0.3631
p=0.8708
p=0.8754
p=0.0344
p=0.2906
p=0.7166
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

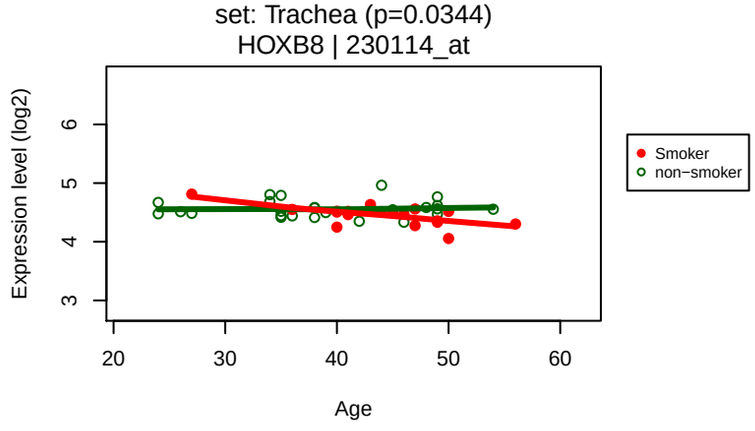

Gene expression ratio in Males
gene: HOXB8 | 230114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1552
p=0.1621
p=0.1931
p=0.5346
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB8 | 230114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4962
p=0.0219
p=0.2569
p=0.767
SmallAirway2
LargeAirway2
Trachea
Macrophage2

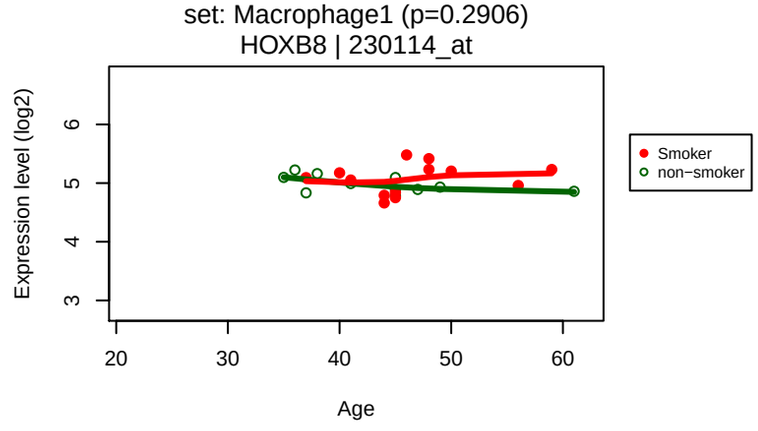

set: SmallAirway1 (p=0.3984)
HOXB8 | 230114_at
Expression level (log2)
Age
Smoker
non−smoker

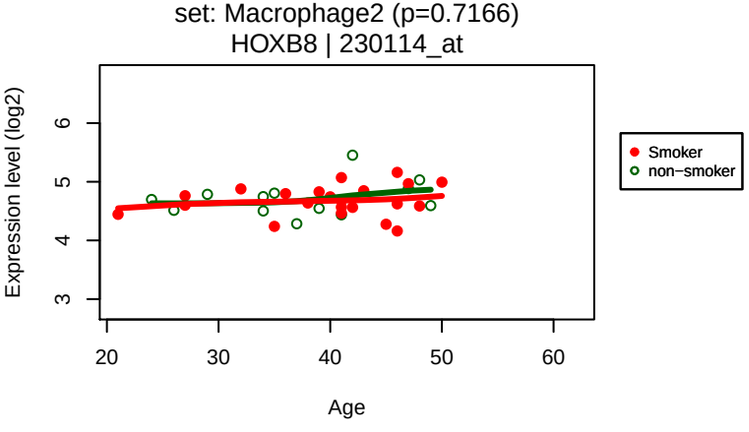

set: SmallAirway2 (p=0.0724)
HOXB8 | 230114_at
Expression level (log2)
Age
Smoker
non−smoker

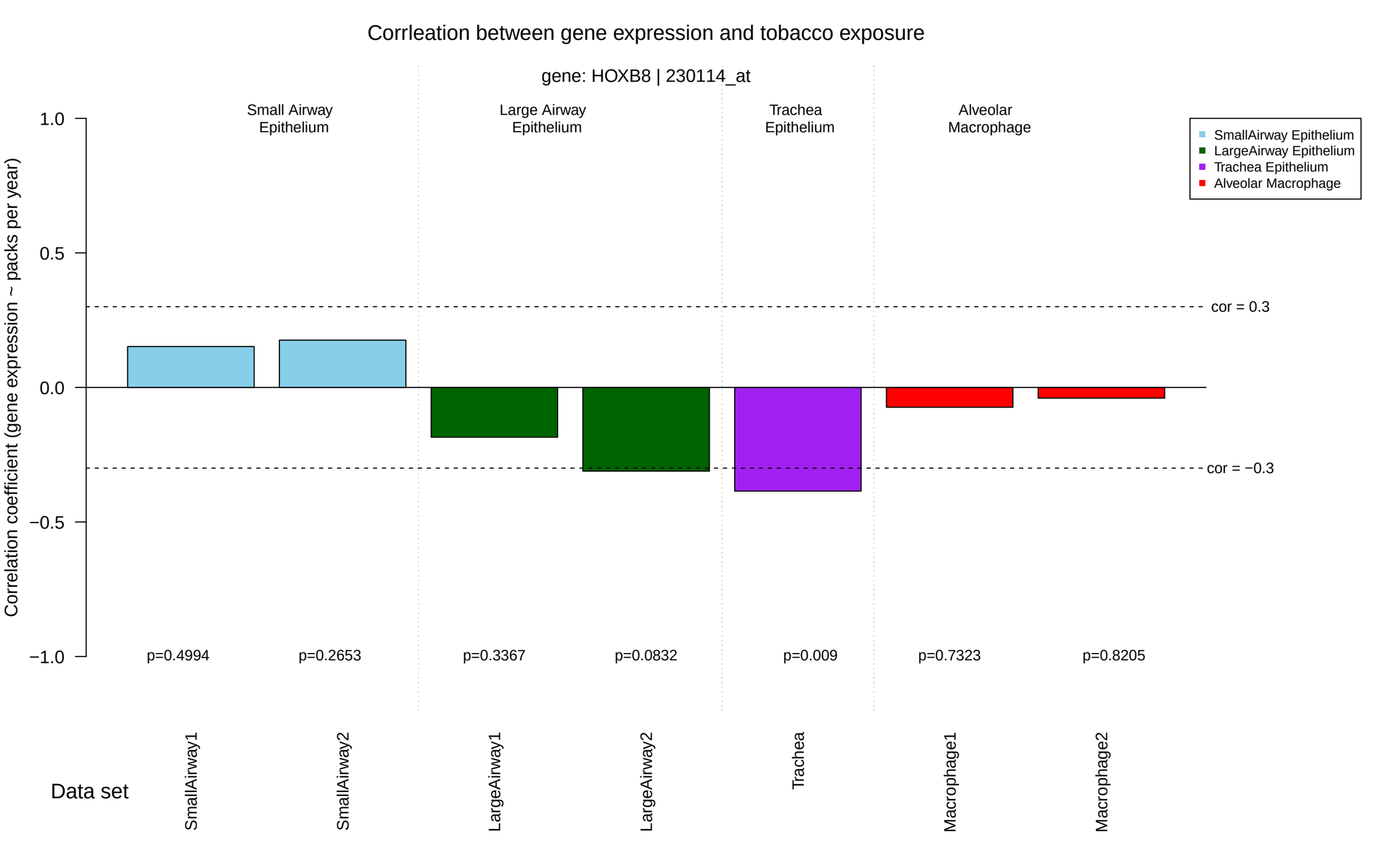

set: LargeAirway1 (p=0.8708)
HOXB8 | 230114_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8754)
HOXB8 | 230114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0344)
HOXB8 | 230114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2906)
HOXB8 | 230114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7166)
HOXB8 | 230114_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB8 | 230114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4994
p=0.2653
p=0.3367
p=0.0832
p=0.009
p=0.7323
p=0.8205
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2